ANSH SINGH

B.Tech. Industrial Internet of Things, 2026 · CGPA 8.45 · New Delhi, India

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SUMMARY

B.Tech. graduate in IoT (CGPA 8.45, 2026) with hands-on experience in Data analytic and AI/ML engineering. Worked as AI Engineer at Gumlabs, where I built AI/NLP features for a voice-keyboard app used by over 400K users, and as Data Analyst Intern at IBM CSR Skillsbuild , automating data pipelines and dashboards. Have independently built and deployed end-to-end projects covering anomaly detection, LLM pipelines, agentic web research, and model monitoring using Python, PyTorch, FastAPI, Docker, and CI/CD. Authored an IEEE conference paper accepted at ICIC3S-2026. Available immediately for full-time roles.

TECHNICAL SKILLS

Languages	Python, SQL,
Generative AI	LangChain, RAG, Prompt Engineering,Transformer architecture ,fine tuning ,LoRA, QLoRA, OpenAI API, Anthropic API, Gemini API, OpenRouter, FAISS (Vector Search), ChromaDB
Machine Learning	PyTorch, TensorFlow, Scikit-learn, Deep Learning, CNNs, NLP, Autoencoders, Anomaly Detection
Backend & MLOps	FastAPI, REST APIs, Web Scraping, JWT Authentication, Docker, GitHub Actions (CI/CD), Vercel, Git
Data & Analytics	PostgreSQL, Power BI ,MS Excel, Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis

EXPERIENCE

Gumlabs Pvt. · Sep 2025 – Jun 2026

- Contributed to Twik AI Voice Keyboard, an AI-powered consumer application scaled to **400K+ users**, working on AI feature development and production pipeline integration.
- Integrated AI/NLP pipelines for voice-to-text workflows, applying speech-processing and natural-language processing techniques to support intelligent text generation and user interactions.

IBM · Data Analyst Intern · Jun 2025 – Aug 2025

- Automated data cleaning and EDA pipelines in Python and SQL across three business datasets, cutting data inconsistencies by 30%.
- Built dashboards adopted for stakeholder reporting, cutting manual reporting time by 40%.

PROJECTS

CardioSense AI · Real-Time Cardiac Anomaly Detection Platform · github.com/Martian172/cardiosense_ai

- Trained an autoencoder in PyTorch to detect cardiac anomalies in ECG signals, reaching 96% accuracy on held-out test data.
- Built a FastAPI backend with JWT authentication and WebSocket streaming for live ECG inference with a React 18 frontend; containerized the stack with Docker Compose and automated deployment through GitHub Actions CI/CD.
- Integrated a Gemini-powered RAG chatbot that explains model predictions in plain language to non-technical users.

LLM Chain Forge · LLM Pipeline Framework with Cost Tracking · github.com/Martian172/llm-chain-forge

- Built a framework from scratch for multi-step LLM pipelines with pluggable OpenAI and Anthropic providers, LRU/TTL caching, and per-call token and cost tracking.
- Shipped a statistical prompt A/B testing harness using the Mann-Whitney U test to compare prompt variants on measurable output quality.
- Included a web playground and CLI; a mock provider makes the full suite offline-testable at zero API cost in CI.

Company Research AI · AI Research Assistant with Web Crawling · companylens-ai.vercel.app · code

- Built an typescript web app agent that takes a company name or URL, crawls the company website, and returns a structured profile with products, AI-generated pain points, and competitors, plus a one-click PDF report.
- Routed AI analysis through OpenRouter so any hosted model can be selected at run time, with schema-validated output, a corrective retry, and an automatic fallback across 8 models when free-tier limits are hit.
- Integrated a Discord bot that auto-posts each finished report and PDF to a channel, with per-error messages so a delivery failure never fails the research run.

MLOps Sentinel · Production Model Monitoring & Alerting Toolkit · github.com/Martian172/mlops-sentinel

- Implemented data-drift detection using the Kolmogorov-Smirnov test and Population Stability Index, with Slack and email alerting on threshold breach.
- Exposed Prometheus metrics and a live WebSocket dashboard; packaged as a one-command Docker Compose stack with Grafana.

RESEARCH PAPER

Physics-Informed Neural Networks with Federated Learning for Microgrid Management · Accepted

ICIC3S-2026, IIIT Una — IEEE

- Proposed a physics-constrained federated learning framework enforcing Kirchhoff's laws in the loss function, enabling edge training without sharing raw data.
- Delivered a 10x inference speedup (0.019s to 0.002s) and 50%+ faster reinforcement learning convergence while preserving full data privacy.

EDUCATION

B.Tech., Industrial Internet of Things · USAR, GGSIPU, New Delhi · CGPA: 8.45 / 10 · Nov 2022 – Jun 2026

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Reinforcement Learning, DBMS

LEADERSHIP

- **Core Organiser, ICAPIE Research Conference** — coordinated speakers and paper submissions across 15+ institutions.
- **Class Representative, USAR (2023–2026)** — sole student-faculty liaison for 120+ peers on scheduling and grievances.